A Low-Cost Single-Channel EEG Seizure Alarm: Code-Faithful Evaluation and a Hardware-Robust Embedded Detector

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***Abstract*—Low-cost amplifiers and Wi-Fi microcontrollers make home EEG seizure alarms easy to build, but their shipped detection code is rarely tested on annotated EEG. We built an open-source single-channel (FP1–FP2) monitor from an EXG BioAmp Pill and an ESP32 and evaluated its code by *code-faithful replay*: CHB-MIT recordings (23 subjects, 194 seizures, 220 h) are converted exactly to the device montage, passed through a front-end model and fed to the unmodified analyzer and a line-by-line firmware port under a simulated clock. The shipped analyzer detected 5.7% of seizures at 0.32 false alarms per hour (FA/h); the firmware's 75.3% at 18.29 FA/h was largely explained by its alarm rate (chance 56.6%). Normalizing the deviation signal the firmware already computes by its own delayed running median (DEV-BR) detected 52.1% (95% CI 36–74%) at 0.97 FA/h on held-out subjects, outperforming the analyzer in 21 of 23 subjects and underperforming in none. With frozen parameters it held 50.0%–52.1% across gains of 500–2000, op-amp clipping, a reduced ADC range and a slower loop, whereas the firmware ranged 7.36–24.37 FA/h. Learned classifiers did not consistently beat this 2.5-KB rule. The result is a verified firmware-only upgrade, not yet deployed and below clinical performance; no bench or human recordings are reported.**

***Index Terms*—seizure detection, wearable EEG, consumer health, ESP32, embedded signal processing, reproducibility**

I. Introduction

Epilepsy affects about 50 million people [1], and unwitnessed nocturnal convulsive seizures are a major risk factor for sudden death [2]. Clinically endorsed wearables sense movement or autonomic signals and reach 90–96% sensitivity at 0.2–0.67 false alarms per day for tonic–clonic seizures [3]; non-convulsive seizures appear only in the EEG. Open front-ends [4] and Wi-Fi microcontrollers [5] make a single-channel EEG alarm cheap to build, but such devices ship hand-tuned thresholds whose code is rarely tested as written. Baseline normalization of line-length is known to help detection [6], and event-based scoring with tolerances is now standard [7]. We contribute: (1) a built FP1–FP2 prototype with ESP32 firmware, desktop analyzer and offline Wi-Fi alerts; (2) code-faithful replay of its unmodified code with subject-level validation and an empirical null; (3) a diagnosis of why the shipped detectors fail; and (4) DEV-BR, a firmware-only detector on an existing signal, robust in simulation to front-end non-idealities and implemented in 2.5 KB of verified C++ (not yet deployed).

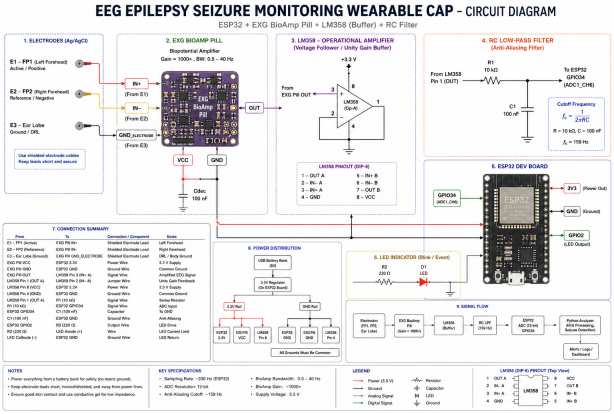


Fig. 1. Prototype: Ag/AgCl electrodes (FP1+, FP2−, earlobe ground), BioAmp Pill, LM358 follower, RC filter, ESP32 ADC. Gain and pass-band are nominal, not measured.

TABLE I
Implementation Status.

Item	Specification	Status
Front-end	BioAmp Pill (≈1000, 0.5–40 Hz, 3.3 V), LM358 follower, RC	built; nominal; 5 V documented [4]
ADC	12-bit, default attenuation, `delay(5)` loop	firmware; rate not measured
Alerts	LED, GPIO pulse, Wi-Fi AP REST API, app	firmware; app not evaluated
Detectors	firmware state machine; 3-feature analyzer	replay-evaluated
DEV-BR	C++ header, 2.5 KB RAM	verified offline; not deployed
Bench data	noise, battery, cost	not measured

II. The Monitoring System

One forehead channel (Fig. 1) is amplified by the BioAmp Pill [4], buffered by an LM358 and digitized by the ESP32 (Table I). The LM358 is not rail-to-rail; at 3.3 V its output tops out near 1.8 V, about 150 mV above mid-supply [8].

Shipped detectors. The firmware tracks a baseline $b[n]{=}0.99b[n{-}1]{+}0.01x[n]$ and deviation $d[n]{=}|x[n]$ –

$\lfloor b[n] \rfloor|$, with an exponentially weighted mean μ and spread σ (factor 0.995 per sample). Thresholds $\mu{+}2.5\sigma$ (BROWNOUT) and $\mu{+}4\sigma$ (BLACKOUT), clamped to fixed ADC ranges, are reset every 30 s, and leaky counters confirm states. A Python analyzer computes line-length (LL), delta/alpha ratio (DAR) and 30–45-Hz energy (HFE) over 2.56 s every 20 samples and requires two of three features above adaptive thresholds with fixed floors on 7 of 8 ticks.

III. Code-Faithful Replay

We used CHB-MIT [9], [10]: all seizure recordings plus one seizure-free file per case; chb01 and chb21 are one subject. FP1–FP2 is reconstructed exactly as the signed sum of the bipolar chain FP1-F7 through FP2-F8; five files lack it, leaving 160 recordings and 194 seizures. Signals pass a causal 0.5–40-Hz model, are resampled to 200 Hz, scaled by gain 1000 around 1.65 V and quantized to 12 bits. The analyzer is imported *unmodified* and driven by a simulated serial port and clock (5 ms per sample); the firmware `loop()` is ported line by line. Following SzCORE [7], alarms within 90 s merge, a seizure is detected by an alarm in [onset−30 s, offset+60 s], and all others are false alarms. CIs use a subject bootstrap, comparisons the Wilcoxon test over 23 subjects, and chance levels 1000 circular shifts of each alarm train.

IV. Why the Shipped Detectors Fail

The analyzer's BLACKOUT alarm detected 5.7% at 0.32 FA/h; the firmware reached 75.3% at 18.29 FA/h, but its shifted alarms already reach 56.6% (Table II). Three causes emerge. (i) *Short memory*: the firmware's spread estimate forgets within about 1 s, so brief 4σ excursions are routine. (ii) *Uninformative votes*: LL (AUC 0.70) and d (0.70) separate ictal from inter-ictal epochs, but DAR is at chance (0.51) and HFE below it (0.40), so the 2-of-3 vote waits on features without seizure information. (iii) *Absolute units*: fixed floors make detection gain-dependent (BLACKOUT 0.0%, 5.7%, 13.4% at gains 500, 1000, 2000).

V. DEV-BR: A Delayed-Median Ratio Detector

DEV-BR addresses all three on the deviation d the firmware already computes. With t in seconds,

$$\delta(t) = \text{mean}\{d[n] : n \in \text{second } t\}, \quad (1)$$

$$\beta(t) = \text{median}\{\delta(\tau) : t{-}329 \le \tau \le t{-}30\}, \quad (2)$$

and an alarm fires when $\delta \ge r\beta$ for D consecutive seconds (90-s suppression, none before 330 s). The 30-s lag keeps the onset out of its own baseline, and the ratio removes the gain. (r, D) is chosen by leave-one-subject-out cross-validation (LOSO): on 22 subjects we take the most sensitive setting meeting an FA/h target and apply it to the 23rd. A nested variant that also selects the signal (LL, DEV or fused) chose DEV in 69 of 69 folds. A header-only C++ class (330-value ring, `nth_element` median once per second, 2.5 KB RAM) reproduced every alarm of the Python evaluation in 800 of 800 runs.

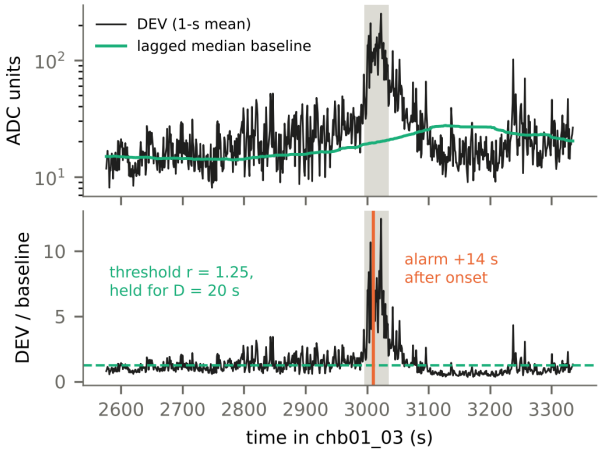


Fig. 2. DEV-BR on a seizure in chb01 (shaded): per-second deviation and delayed median baseline (top), their ratio, threshold and alarm (bottom).

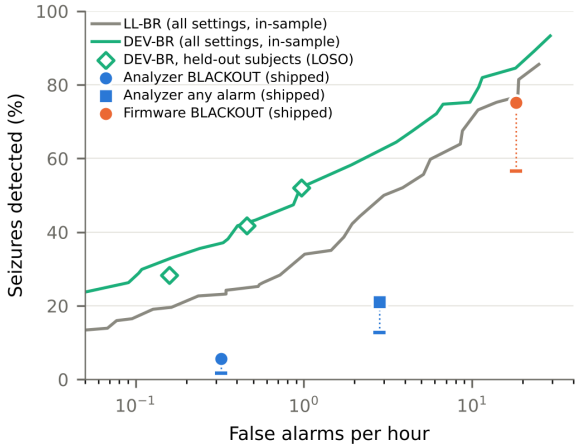


Fig. 3. Sensitivity versus FA/h on FP1–FP2. Lines: best in-sample ratio trade-offs. Diamonds: DEV-BR on held-out subjects. Markers: shipped detectors; dashes: their chance levels.

Results. On held-out subjects DEV-BR detected 101/194 seizures (52.1%, CI 36–74%) at 0.97 FA/h with median latency 22.0 s, and 41.8% at 0.46 FA/h (Figs. 2, 3). That is about 7 times the analyzer at a comparable alarm rate: better in 21 of 23 subjects, worse in 0 ($p{=}6{\times}10^{-5}$); its chance level was 6.5%. Without normalization the same LOSO procedure reached only 32.5%; removing the lag or lengthening the window to 600 s also hurt (Table III). Excluding the 15 seizures that begin during warm-up gives 54.2%.

Simpler is enough. k-of-n and dual time-scale rules, a baseline carried across files, logistic regression and gradient-boosted trees on 14 baseline-normalized features (thresholds by inner cross-validation) reached at most 51.0% at $\le$1 FA/h. Boosted trees gave 44.3% versus 41.8% at $\le$0.5 FA/h, a non-significant difference ($p{=}0.25$). Averaging or OR-ing ratios from the neighbouring frontal derivations FP1-F7 and FP2-F8 did not help either. The limit is the frontal recording site, not the rule.

Hardware robustness. With parameters frozen, DEV-BR

TABLE II
EVENT-BASED RESULTS ON FP1–FP2 (23 SUBJECTS, 194 SEIZURES, 220 H). BRACKETS: SUBJECT-BOOTSTRAP 95% CI. "CHANCE": SENSITIVITY OF THE CIRCULARLY SHIFTED ALARM TRAIN. BR DETECTORS: HELD-OUT SUBJECTS ONLY (LOSO).

Detector	Detected	Sensitivity [95% CI]	Macro	FA/h	Latency (s)	Chance
Analyzer, BLACKOUT (shipped)	11/194	5.7% [2–12]	10.0%	0.32	26.3	1.6%
Analyzer, any alarm (shipped)	41/194	21.1% [14–29]	25.4%	2.82	28.4	12.8%
Firmware, BLACKOUT (shipped)	146/194	75.3% [60–94]	87.6%	18.29	14.5	56.6%
LL-BR, $\leq$1 FA/h	49/194	25.3% [15–42]	40.1%	1.04	32.0	4.8%
FUSE-BR, $\leq$1 FA/h	78/194	40.2% [24–64]	56.0%	0.96	21.5	–
DEV-BR, $\leq$1 FA/h	101/194	52.1% [36–74]	68.6%	0.97	22.0	6.5%
DEV-BR, $\leq$0.5 FA/h	81/194	41.8% [26–65]	59.5%	0.46	23.0	3.9%
DEV-BR, $\leq$0.2 FA/h	55/194	28.4% [15–49]	46.1%	0.16	29.0	–

TABLE III
ABLATIONS AND ALTERNATIVE DETECTORS (LOSO, FP1–FP2): SENSITIVITY / FA/H AT THE $\leq$1 AND $\leq$0.5 FA/H TARGETS.

Variant	$\leq$1 FA/h	$\leq$0.5 FA/h
reference (300 s, lag 30 s, median)	52.1% / 0.97	41.8% / 0.46
no lag (0 s)	50.5% / 0.79	34.5% / 0.42
lag 10 s	51.0% / 0.87	40.7% / 0.45
lag 60 s	47.9% / 0.93	38.7% / 0.46
window 60 s	50.5% / 0.94	43.3% / 0.45
window 120 s	53.1% / 0.88	38.1% / 0.43
window 600 s	41.2% / 0.94	34.0% / 0.47
mean instead of median	50.0% / 0.78	41.2% / 0.35
no normalization (absolute)	32.5% / 0.83	30.4% / 0.42
k-of-n persistence	43.3% / 0.95	36.6% / 0.38
dual time-scale rule	51.0% / 1.01	42.8% / 0.49
baseline carried across files	48.5% / 0.97	41.8% / 0.46
log. regression, firmware features	43.3% / 1.03	35.1% / 0.44
log. regression, all features	43.8% / 0.90	38.7% / 0.47
boosted trees, all features	47.9% / 1.01	44.3% / 0.50

TABLE IV
FRONT-END ROBUSTNESS (SIMULATED). "FIXED": DEV-BR WITH r=1.25, D=20 S CHOSEN AT GAIN 1000. CLIP: LM358 1.8-V CEILING; SAT.: SATURATED SAMPLES. CELLS: SENSITIVITY / FA/H.

Condition	Sat.	DEV-BR fixed	DEV-BR LOSO	Firmware
$\times$500	0.00%	52.1% / 0.98	47.9% / 0.94	51.5% / 7.36
$\times$1000 (ref.)	0.00%	52.1% / 0.95	52.1% / 0.97	75.3% / 18.29
$\times$2000	0.01%	52.1% / 0.97	50.0% / 0.96	86.1% / 24.37
$\times$1000, LM358 clip	0.52%	51.5% / 0.97	48.5% / 0.96	73.2% / 18.01
$\times$2000, LM358 clip	2.69%	50.0% / 0.93	50.0% / 0.93	82.0% / 23.09
$\times$1000, 3.1-V ADC	0.00%	52.1% / 0.96	50.5% / 0.98	78.9% / 19.22
$\times$1000, 167-Hz loop	0.00%	51.5% / 0.95	51.5% / 0.96	45.9% / 14.74

stayed at 50.0%–52.1% and 0.93–0.98 FA/h across gains, LM358 clipping (2.7% of samples saturated at gain 2000), a 3.1-V ADC range and a 6-ms loop, while the firmware moved between 45.9%–86.1% and 7.36–24.37 FA/h (Table IV).

VI. DISCUSSION AND CONCLUSION

Replaying the code that ships showed that the prototype's on-device alarm mostly fires at random, and that a decision rule on a signal it already computes makes it a significant, hardware-robust detector at about one false alarm per hour, with no added hardware. Three principles generalize to low-cost biosignal devices: validate the shipped code, not a re-implementation; normalize against the wearer's own delayed background so decisions are invariant to front-end tolerances; and stop adding complexity once held-out tests show the signal is the bottleneck. The absolute performance is far from clinical wearables [3], and the device is a research prototype. Limitations: results come from software replay of a pediatric clinical dataset with a nominal front-end model; non-idealities were simulated, not measured; the BioAmp runs below its documented supply; recordings start cold; and DEV-BR is not yet on the device. Next steps are bench characterization, on-device deployment, a rail-to-rail buffer, a temporal or behind-the-ear channel and an ethics-approved recording study. Hardware, firmware, analyzer, replay benchmark and C++ detector are open source [11].

ACKNOWLEDGMENT

The authors used an AI assistant (Claude, Anthropic) to help develop the evaluation code and draft the manuscript, and take full responsibility for the content.

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